

Instructions

- All questions carry equal weightage, except simulation questions, which have double the weightage.
- Copying will result in a loss of marks/penalty.
- All assumptions must be clearly stated along with appropriate figures (if any); otherwise, grading will not be done.
- Including the source code for simulation questions, along with the results, is mandatory. The source code must be properly commented, explaining each line.

1. Two point charges of equal magnitude q are positioned at $z = \pm d/2$. We aim to determine:

- (a) The electric field everywhere on the z -axis.
- (b) The electric field everywhere on the x -axis.
- (c) The results of (a) and (b) if the charge at $z = -d/2$ is $-q$ instead of $+q$.

2. A crude device for measuring charge consists of two small insulating spheres of radius a , one of which is fixed in position. The other is movable along the x -axis and is subject to a restraining force kx , where k is a spring constant. The uncharged spheres are centered at $x = 0$ and $x = d$, the latter fixed. If the spheres are given equal and opposite charges of Q/C , obtain the expression by which Q may be found as a function of x . Determine the maximum charge that can be measured in terms of ϵ_0 , k , and d , and then state the separation of the spheres. What happens if a larger charge is applied?

3. A flux density field is given as

$$\mathbf{F}_1 = 5\mathbf{a}_z.$$

The task is to evaluate the outward flux of $\mathbf{F}_1$ through the hemispherical surface defined by $r = a$, $0 < \theta < \pi/2$, and $0 < \phi < 2\pi$. Next, consider what simple observation would have saved a lot of work in the previous part. Identifying symmetries or using alternative methods can simplify the calculations significantly. Now suppose the field is given by

$$\mathbf{F}_2 = 5z\mathbf{a}_z.$$

Using the appropriate surface integrals, determine the net outward flux of $\mathbf{F}_2$ through the closed surface consisting of the hemisphere from the first part and its circular base in the xy plane. Finally, repeat the previous calculation by applying the divergence theorem and evaluating an appropriate volume integral. This approach should confirm the result obtained through direct surface integration.

4. An infinitely long cylindrical dielectric of radius b contains charge within its volume with a charge density given by

$$\rho_v = a\rho^2,$$

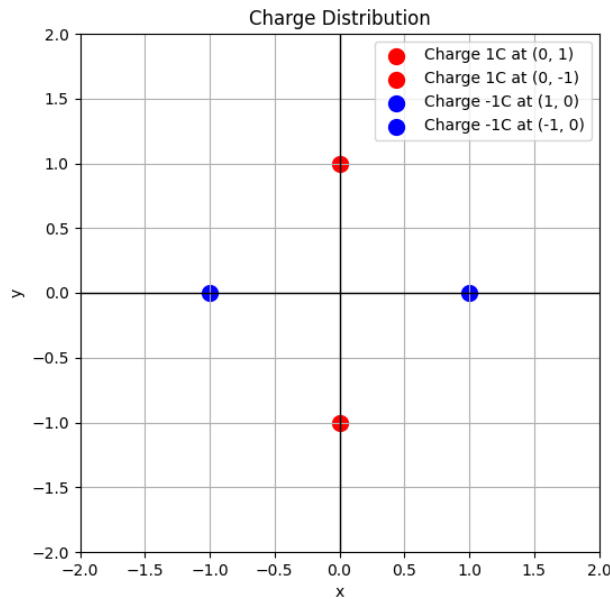
where a is a constant. The goal is to determine the electric field strength $\mathbf{E}$ both inside and outside the cylinder.

5. Given the vector field in cylindrical coordinates:

$$\vec{F} = \left[\frac{40}{s^2 + 1} + 3(\cos \phi + \sin \phi) \right] \hat{s} + 3(\cos \phi - \sin \phi) \hat{\phi} - 2\hat{z}$$

- (a) Compute and plot the magnitude $|\vec{F}|$ as a function of ϕ for $s = 3$.
- (b) Compute and plot the magnitude $|\vec{F}|$ as a function of s for $\phi = 45^\circ$.
- (c) Calculate the divergence $\nabla \cdot \vec{F}$.
- (d) Calculate the curl $\nabla \times \vec{F}$ and verify whether the field is conservative.

6. Refer to the charge distribution below;



- (a) Plot the Electric Field.
 - (b) Plot the potential with magnitude represented along the Z-axis.
 - (c) Compute the potential energy of the configuration.
 - (d) Use Gauss's Law to verify the divergence of the electric field.
 - (e) Verify whether the curl of the electric field is zero.
7. Given the spherically symmetric potential field in free space, $V(r) = V_0 e^{-r/a}$, determine the following:
(a) Find the charge density ρ_v at $r = a$. (b) Calculate the electric field $\mathbf{E}$ at $r = a$. (c) Compute the total charge.
8. A parallel-plate capacitor has plates located at $z = 0$ and $z = d$. The region between the plates is filled with a material that contains a uniform volume charge density ρ_0 C/m³ and has permittivity ϵ . Both plates are held at ground potential.

- (a) Determine the potential field between the plates.
- (b) Determine the electric field intensity $\mathbf{E}$ between the plates.
- (c) Repeat parts (a) and (b) for the case where the plate at $z = d$ is raised to a potential V_0 , with the plate at $z = 0$ grounded.